From RNA-seq Data to Biological Insight:
Data-Informed. Biology-Driven.

A complete analysis should answer the biological question, not only produce plots!

RNA-seq can reveal how expression changes across tissues, treatments, developmental stages, genotypes, or disease states. BiOmixVita connects the statistical result to the biological context needed to decide what to investigate next.

> **A question-led workflow**
> We keep the analytical foundations consistent and reproducible, while adapting comparisons, visualizations, annotations, and interpretation to the design of each project.

RNA-seq analysis often produces several familiar visualizations, but each one answers a different question. PCA plots reveal global sample structure. Heatmaps show coordinated expression patterns across genes and samples. Volcano plots bring together the magnitude of change and statistical significance to highlight genes that are both strongly changed and well supported by the data.

None of these views should be interpreted in isolation. A heatmap pattern does not prove a mechanism, and a highly significant gene is not necessarily biologically important on its own. The strongest interpretation connects global sample structure to gene-level changes, pathway-level patterns, and the experimental context in which the data were generated.

> ***Data-Informed. Biology-Driven.***
> In this white paper, we show how BiOmixVita keeps the analytical foundations consistent and reproducible, while adapting comparisons, visualizations, annotations, and interpretation to the design of each project. The goal is to move beyond familiar outputs and toward the biological understanding that guides the next decision.

Case study: genotype across development

A wild-type line and a gene mutant were profiled at embryo and adult stages. Four comparisons separated genotype effects from normal developmental change. The analysis showed that developmental stage was the dominant source of transcriptional variance, while the mutant created a distinct adult signature worth pursuing.

WT vs mutant · embryo Genotype effect at the embryonic stage	**WT · embryo vs adult** Normal developmental change
Mutant · embryo vs adult Developmental change in the mutant	**WT vs mutant · adult** Genotype effect at the adult stage

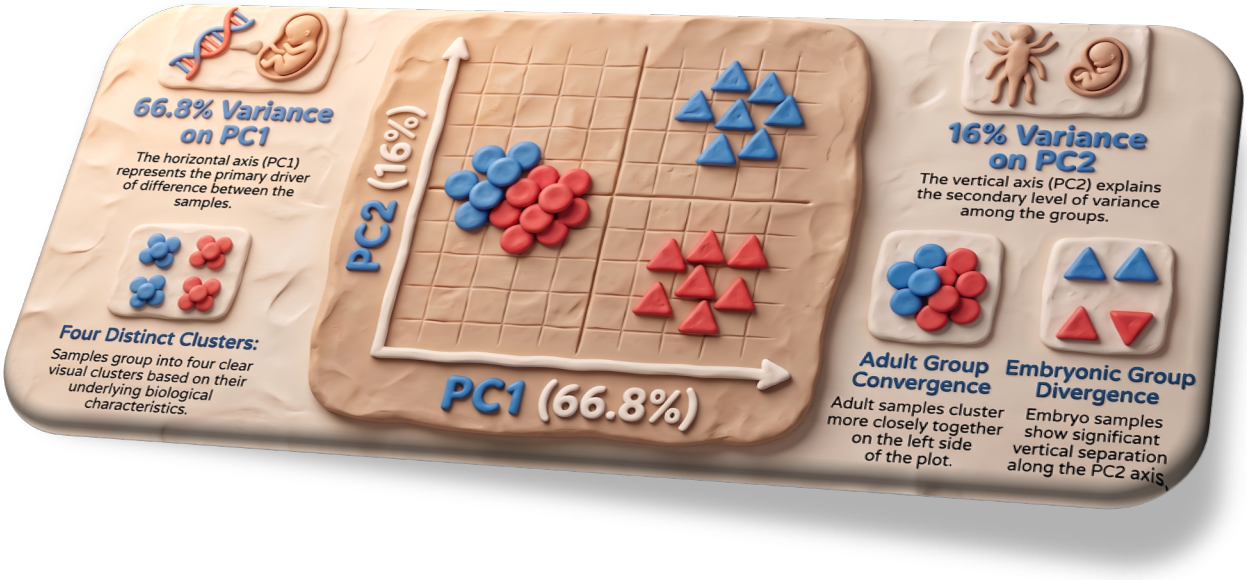
1. **The analytical journey**

01	02	03	04	05
Explore **QC, filtering, sample structure**	Normalize **Comparable expression values**	Test **Differential expression, FDR control**	Interpret **GO, KEGG, GSEA, pathways**	Deliver **Interactive report + data package**

2. Explore gene expression patterns

Before testing individual genes, we establish whether the data reflect the experimental design. PCA, correlation views, and heatmaps reveal replicate consistency, group structure, outliers, and possible sources of unwanted variation.

BiOmixVita Output:



Stage and genotype are made visible together!

a. What the figure answers

The principal-component axes are not treated as a fixed template. **PC1, PC2, PC3, and subsequent components are reviewed, and the most informative combinations are selected for the clearest group clustering and biologically meaningful visualization.** Labels, shapes, and annotations then reveal the factors that define the experiment. The result says why samples separate, not just that they do. [1]

b. BiOmixVita output

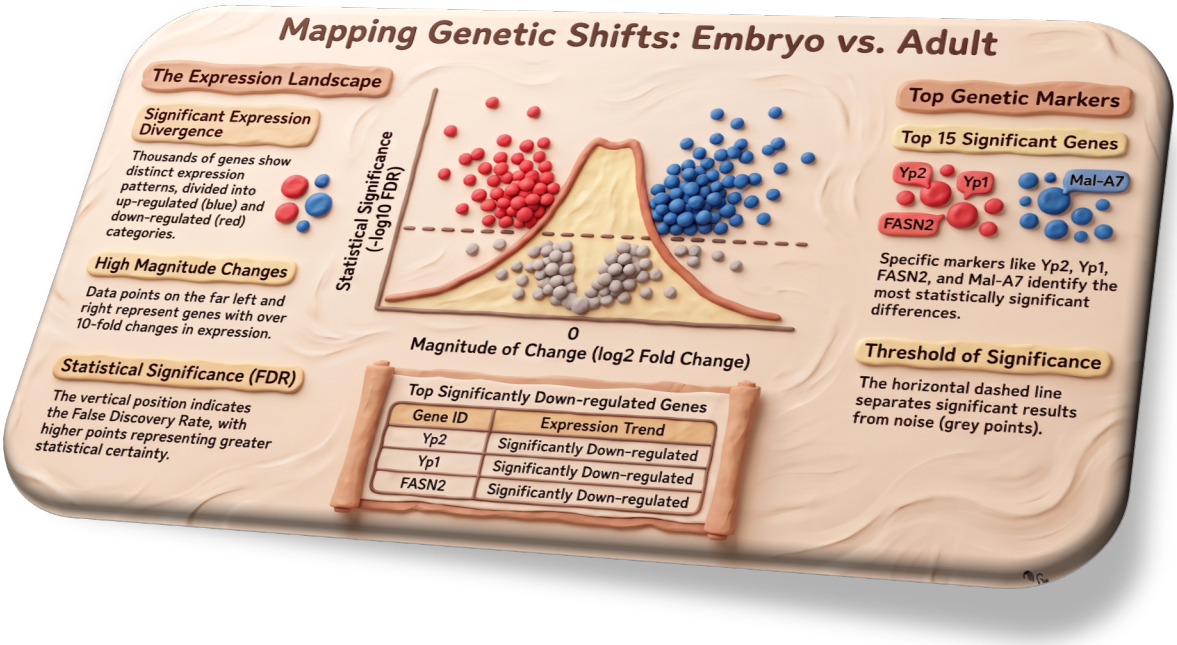
BiOmixVita exploratory analysis selects the component views that best expose structure between groups, while preserving the context needed to understand how samples relate globally and how every downstream comparison should be read.

In this case, the dominant axis separated embryo from adult samples. Replicates clustered tightly, with no obvious outliers. Selecting the clearest PC combination made that structure easier to see and interpret, rather than presenting PC1 and PC2 simply because they are the default axes.

3. Identify differential expression

Differential expression quantifies how genes change between defined conditions. We combine effect size, statistical significance, replicate structure, and project-specific contrasts so the result is interpretable rather than a list of isolated hits. [3]

BiOmixVita Output:



a. What the figure answers

BiOmixVita applies documented FDR and effect-size thresholds, then labels top genes by both criteria with adaptive placement for a publication-ready view. [3]

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b. BiOmixVita output

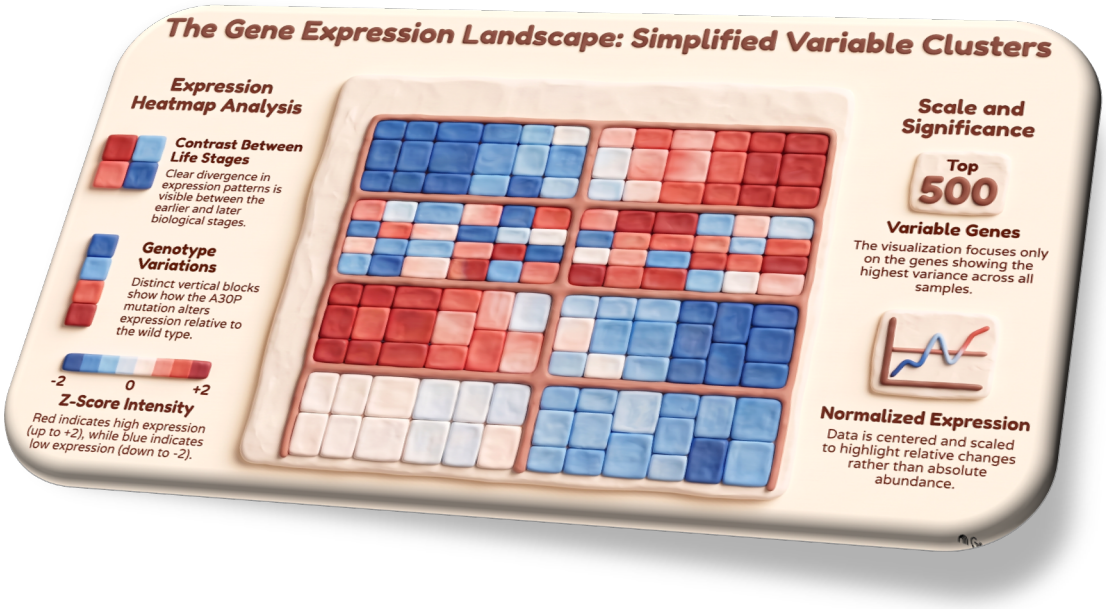
The output connects each significant result to the comparison, direction of change, magnitude of effect, and supporting evidence.

The goal is not to decorate the volcano plot. It is to make the decision boundary visible and to keep the genes a scientist would discuss in the room readable on the page.

4. Resolve coordinated expression programs

Genes rarely act alone. Clustering and heatmap analysis identify groups of genes that move together across samples, helping separate shared biological programs from condition-specific responses. [2]

BiOmixVita Output:



Clusters become defined, named, and testable expression programs!

a. What the figure answers

BiOmixVita tunes clustering to the design, annotates the resulting programs, and labels representative driver genes so the pattern becomes a testable hypothesis.

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b. BiOmixVita output

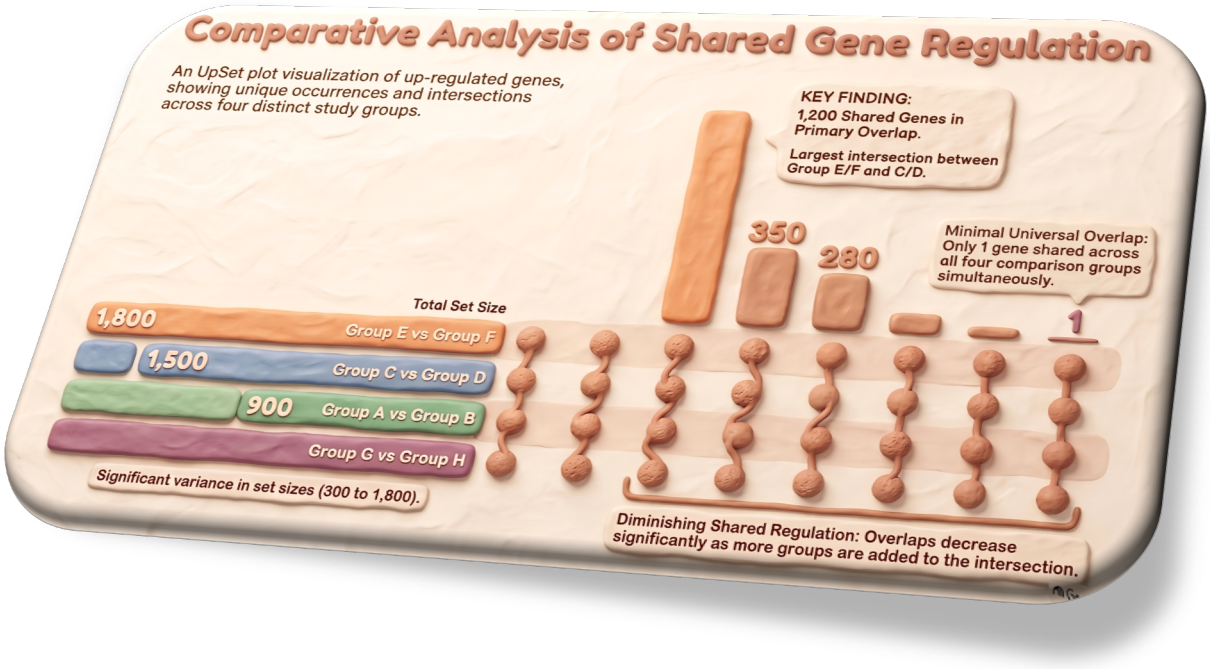
The output turns a high-dimensional expression matrix into interpretable blocks linked to the experimental question.

In the case study, the variable-gene landscape separated a shared developmental program from an adult-mutant signature. That distinction is more actionable than a long list of significant genes.

5. **Compare responses across conditions**

Multiple contrasts can show what is shared, what is condition-specific, and what changes only in a particular tissue, genotype, or stage. UpSet-style views keep those relationships traceable across the full design.

BiOmixVita Output:



Every intersection remains traceable to genes!

a. **What the figure answers**

BiOmixVita separates up- and down-regulated sets, scales across comparisons, and provides membership tables so every intersection can be queried.

b. **BiOmixVita output**

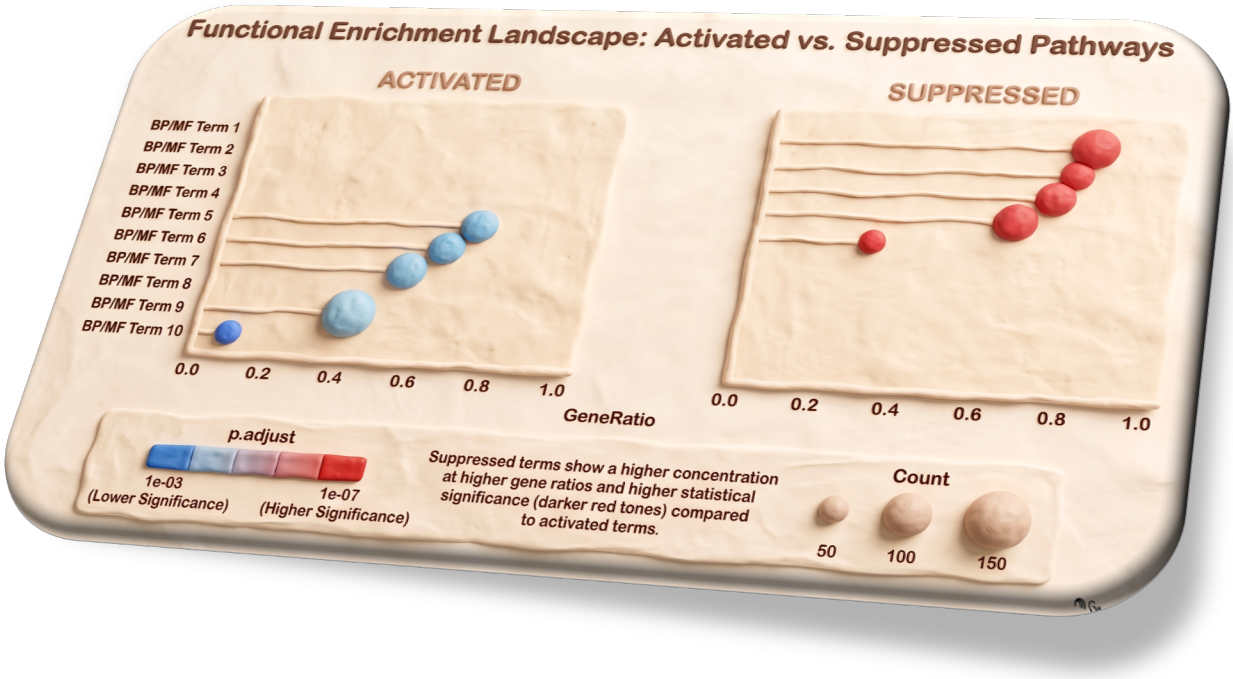
The output helps distinguish core responses from context-dependent biology.

The output is not only a picture of overlap. It is a route back to the genes that are unique to one contrast, shared by two, or common across the entire experiment.

6. Translate gene changes into pathways

Pathway analysis places differential expression into a broader biological context. We use ranked enrichment and direction-specific views to capture coordinated changes, then connect enriched terms to the genes and pathways behind them.

BiOmixVita Output:



Full-ranked enrichment keeps coordinated shifts in view!

What the figure answers

GSEA uses the ranked distribution, separates direction, reports broad and specific terms, and can overlay gene-level expression on the pathway map. [4]

BiOmixVita output

The result moves from “which genes changed?” to “which biological processes and mechanisms are shifting?”

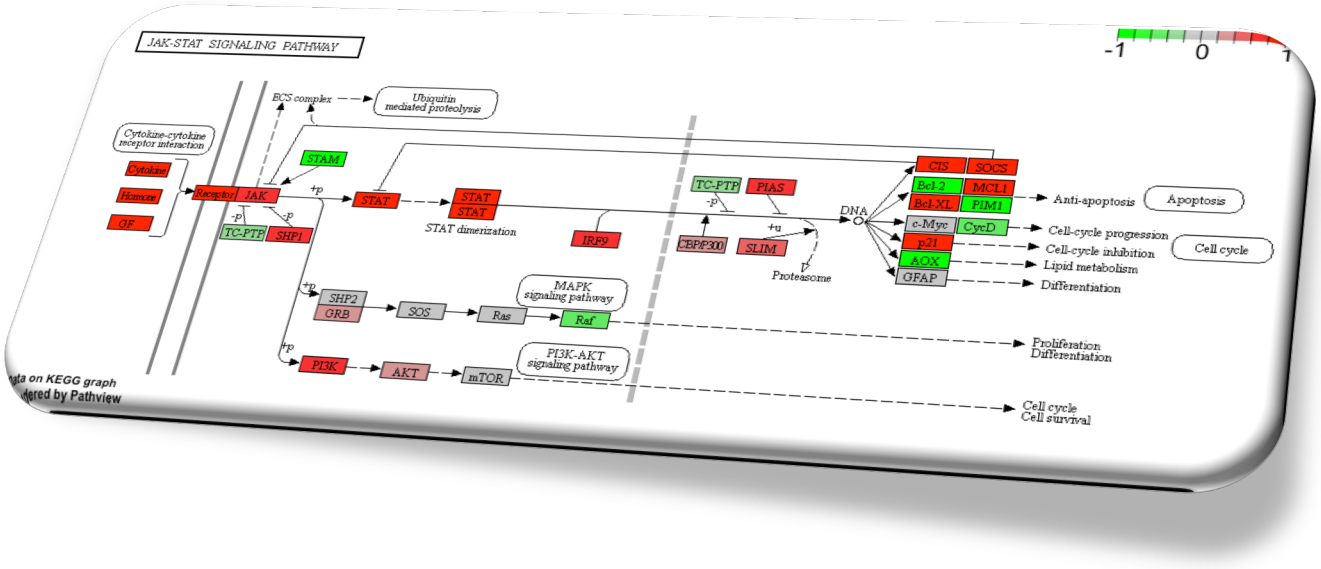
This is where “a pathway is enriched” becomes a mechanistic hypothesis: which genes moved, in what direction, and where in the pathway should the next experiment look?

06 / Visualize mechanisms and regulatory context

When the question moves beyond enrichment, pathway maps and interaction networks show where genes connect, which modules are active, and which regulatory relationships may explain the observed phenotype. [6]

Illustrative example: *This KEGG pathway image is shown randomly as an example from a published course analysis available on* ***Rpubs****. It is not derived from, and is not related to, any analysis project performed by the BiOmixVita team.* [9]

BiOmixVita Output:



Affected pathway views help translate molecular changes into the next biological question!

What the figure answers

BiOmixVita can adapt KEGG pathway views to the tissue and comparison, overlaying gene-level changes so the biological mechanism becomes easier to inspect and communicate. [5]

BiOmixVita output

The output places pathway-level evidence inside the wider report, alongside the QC, differential expression, statistics, and reproducibility information needed to evaluate it.

The pathway view sits inside the wider report, alongside the evidence behind it: QC, differential expression, enrichment statistics, annotated tables, and reproducibility information.

7. **Analysis Deliverables: A complete RNA-seq data outputs, ready to use.**

Every engagement is delivered as a self-contained analysis package. The result is organized to move from exploratory structure to differential expression, pathway interpretation, and the next research decision.

Interactive HTML report

QC, normalization, clustering, differential expression, and enrichment in one navigable document.

QC & normalization figures

Library size, normalization factors, dispersion, mean–variance, and before/after expression views.

Comparison-ready results

PCA, correlation heatmaps, variable-gene heatmaps, volcano plots, MA plots, and multi-comparison overlap.

Functional interpretation

GO and KEGG enrichment, direction-specific views, and pathway overlays for the biology that matters to the project. [5] [8]

Processed data

Normalized CPM and log-CPM matrices, full and significant-only tables, and analysis-ready files.

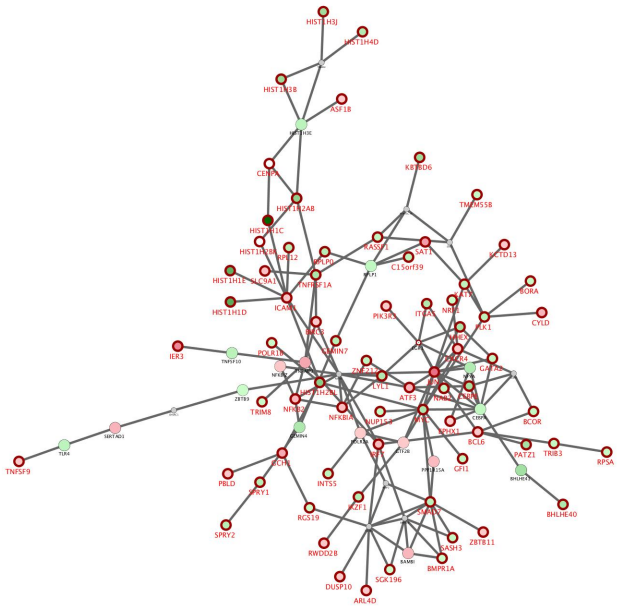
Methods & reproducibility

Pinned software and database versions, parameters, thresholds, and a version-controlled R environment.

Optional extension

Add-on analyses and visualization layers are designed on a project-by-project basis. For example, miRNA-seq profiling can be performed in parallel with RNA-seq, connecting small-RNA regulators to differentially expressed mRNA targets through an integrated regulatory network. We can construct, filter, and visualize gene and pathway networks to identify densely connected modules, highlight hubs and communities, and explore candidate regulatory mechanisms using network-analysis tools. [6] [7]

This is only one example of a possible extension. Depending on the scientific question, the integration may also include proteomic, metabolomic, phenotypic, clinical, imaging, or other quantitative measurements that the project requires. These modalities can be jointly analyzed and visualized within a customized network-based framework. Every extension is tailored to the project design, available data, and biological objectives, **100% customized to the customer's research question.** [10]



DGE-Integrative Network: connecting, mRNA-seq, with other omics or metrics data.

8. **Rigor, Reproducibility & Transparency**

Established methods, documented thresholds, organism-matched annotations, and version-pinned software keep every result statistically rigorous, auditable, and reproducible.

References

The references below support the analytical methods, databases, and network-analysis concepts discussed in this paper. The KEGG image used in the pathway section is explicitly identified as an external illustrative example, not as a BiOmixVita project result.

[1] Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. Philosophical Transactions of the Royal Society A, 374(2065), 20150202. https://doi.org/10.1098/rsta.2015.0202

[2] Eisen, M. B., Spellman, P. T., Brown, P. O., & Botstein, D. (1998). Cluster analysis and display of genome-wide expression patterns. Proceedings of the National Academy of Sciences, 95(25), 14863–14868. https://doi.org/10.1073/pnas.95.25.14863

[3] Love, M. I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biology, 15(12), 550. https://doi.org/10.1186/s13059-014-0550-8

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[6] Shannon, P., et al. (2003). Cytoscape: A software environment for integrated models of biomolecular interaction networks. Genome Research, 13(11), 2498–2504. https://doi.org/10.1101/gr.1239303

[7] Bader, G. D., & Hogue, C. W. V. (2003). An automated method for finding molecular complexes in large protein interaction networks. BMC Bioinformatics, 4, 2. https://doi.org/10.1186/1471-2105-4-2

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[9] Barry Usher. (2018). Class 15: Pathway analysis. RPubs. https://rpubs.com/barryus/class15

[10] BiOmixVita Team. (2026). From Data to Decision: Scalable Biological Intelligence Powered by BiOmixVita. BiOmixVita Insights. https://biomixvita.com/insights/2